Introductory Into PyTorch: Tensors

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Exercises

1. Create the following tensors manually using `torch.tensor`:
   - A 1D tensor with values $[1.0, 2.0, 3.0]$.
   - A 1D NumPy array with values $[1.0, 2.0, 3.0]$, transform it to torch tensor and back to a NumPy array.
   - A 2D tensor with values $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.
2. Create a 2×3 tensor filled with:
   - Zeros.
   - Ones.
   - Random values from $[0, 1)$.
3. Let $a = [1.0, 2.0, 3.0]$ and $b = [4.0, 5.0, 6.0]$. Compute:
   - $a + b$
   - $a \cdot b$ (element-wise product)
   - Dot product of a and b
   - Mean of a
4. Given the matrix $m = \begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \end{bmatrix}$, extract:
   - The first row
   - The second column
   - The element at position (1,2)
5. Create a tensor using `torch.arange(12)`. Then:
   - Reshape it into a 3×4 tensor
   - Flatten it back into a 1D tensor
   - Transpose into a 4×3 tensor.

- Reshape into $2 \times 2 \times 3$ tensor

6. Perform in-place operations:
   - Add 10 to all elements of a tensor
   - Multiply the result by 2

7. Consider a 3D tensor T of shape $(2, 3, 4)$ representing 2 matrices of size 3×4. Perform the following operations:
   - Compute the sum over the last dimension (dim=2), resulting in a 2×3 matrix.
   - Compute the maximum over the second dimension (dim=1), resulting in a 2×4 matrix.
   - Compute the mean over the first dimension (dim=0), resulting in a 3×4 matrix.

8. Let A be a 3×3 matrix and v a vector of size 3. Compute:
   - The matrix-vector product Av.
   - The matrix-matrix product AB, where B is another 3×3 matrix.

9. Create a batch of 2 matrices of shape 3×4 and another batch of 2 matrices of shape 4×2. Use batch matrix multiplication to multiply them and obtain a result of shape $2 \times 3 \times 2$.

10. (Bonus) Consider a function $f(x, y) = x^2 + xy + y^2$. Create a 2D tensor grid of x and y values using `torch.meshgrid`, evaluate f on the grid, and compute the average value of f over the domain.